

London 01/04/21

Dear Dr Rubin,

I am writing this letter to support the application of Aleksandra A. Ziolkowska for a Science Corps fellowship. Aleksandra is a current PhD student at the University of Oxford (Physics Department). Her PhD supervisor is Prof. Essler. However, for the past year she has been working on a project with me, so I consider myself her co-supervisor (even if unofficially). She approached me spontaneously after meeting a few times at conferences and asked if she could work with me on a project of my choice. After getting the go-ahead from her supervisor, we started working together. Despite Covid restrictions, I have had time to see how Aleksandra works and have got to know her quite well over the past few months. We have been meeting on-line once a week and have written a paper together (with two other collaborators). Previous to this, I had also met her at several international conferences and schools and had the opportunity to form some opinion of her ability and character already at that stage.

In terms of her technical ability, I have no hesitation to recommend Aleksandra. She has excellent grades from her BSc and MSc and is now working in a very specialized area of Mathematical Physics, having secured a very competitive PhD scholarship. Her work is concerned with quantum systems, that is matter at very small scales, and in particular, quantum systems out of equilibrium, that is the properties of large groups of atoms when subjected to a sudden change to their environment. This is an area where various kinds of techniques are employed, from advanced mathematics, to programming in order to simulate the quantum systems under consideration and check that the proposed mathematical description is accurate. Aleksandra has done work on both. She has published a paper with her supervisor which is of a rather mathematical nature, whereas the work she has done with me and others was numerical and she, together with another student, led the development of the numerical algorithm. Aleksandra has a slightly unusual background as she started her University education doing Electrical Engineering, did Laboratory work in Singapore for one year, and then has gone on to work in a very mathematical area of Physics. I think this background is an asset in terms of how she can inspire students in her teaching and the breadth of examples she would be familiar with.

During her PhD in Oxford, she has been teaching a small group of students at her college and she seems to enjoy the experience very much and to be successful at it (although I only know this from her account). Her oral communication and presentation skills are excellent, as I had the opportunity to see in one of her recent talks. She recently was awarded a prize for one of her presentations. Her communication skills are also evident when we discuss. She is able to express questions and comments clearly, even when we talk about rather complex phenomena. In discussions, she is able to hold her own, she is confident and resilient. She is also creative and keeps suggesting new ideas, and new problems, usually quite ambitious. She is not easily put off by difficulties. I particularly like this trait in her.

Aleksandra has a strong personality and I feel confident that she would be able to deal with difficult situations if they arise. She is independent and adventurous, curious about the world, new people and new places. I think some of her confidence comes from her childhood. When she was very young (I think around 12) she became a champion dog trainer in Poland and travelled to many competitions in Poland and abroad with her dog. I think this must have taught her self-reliance and confidence. She also travelled with her family from a young age (she has told me about going to Egypt as a teenager) and has since been to many parts of the world, including spending one year in Singapore. Two years ago, we found ourselves at a conference in Brazil. Aleksandra does not speak Portuguese, but she still was able to get what she needed and to complain when things did not go to plan (i.e. at restaurants!). During the conference she was the main person suggesting activities for the rest of us, things to try and see and she was certainly not afraid of exploring the new environment (even if homicide rates at our location were very high and made many other people rather cautious). It was not crazy risk-taking but a healthy curiosity about the place. After the conference she went travelling in Brazil on her own for several weeks.

Aleksandra has the skills and potential to develop a research career and I hope she will do so. It all depends on job opportunities, which are unfortunately scarce in our area. She has been applying for postdoctoral fellowships and hoping to continue working in quantum physics. She sees her work for Science Corps as a transition to postdoctoral life and a valuable experience. As I indicated, she likes new people and new places and she likes teaching, so this is perfect for her. She is also a socially conscious and engaged person, particularly in what concerns gender issues within our community, where female academics are still, unfortunately, scarcely represented. One of the reasons she was interested in working with me is that I am one of the few senior women in my area of research and she has often asked me about my experience of being “the only woman at the conference” and how I have dealt with it. In an ideal world I would see her with a permanent academic position in 10 years, but I know that this is also dependent on a bit of luck and on how her life develops in the next few years (academic careers become a lot more complicated once a partner is also involved, as there is often the need to move to a different country every few years). In the meantime, I hope you will be able to offer her this opportunity to give something back to society.

Yours Sincerely,



Dr Olalla Castro-Alvaredo
Reader in Theoretical Physics
Department of Mathematics
City, University of London
E-mail: sa702@city.ac.uk
Tel.: 020 7040 8952
Web: www.olallacastroalvaredo.weebly.com